



資料結構

Data Structure

bonus

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bonus-Q1.Leetcode

Winter is coming! During the contest, your first job is to design a standard heater with a fixed warm radius to warm all the houses.

Every house can be warmed, as long as the house is within the heater's warm radius range.

Given the positions of houses and heaters on a horizontal line, return *the minimum radius standard of heaters so that those heaters could cover all houses*.

Notice that all the heaters follow your radius standard, and the warm radius will be the same.

Code

```
class Solution {
public:
    int findRadius(vector<int>& houses, vector<int>& heaters) {
        int R = 0;          // 定義半徑
        for (int house : houses) {      // 遍歷 houses vector
            int minD = INT_MAX;          // 將最小距離 minD 賦予 int 的最大值，方便找最小
            for (int heater : heaters) { // 遍歷 heaters
                minD = min(minD, abs(house - heater)); // 找最小距離並更新
            }
            R = max(R, minD); // 更新最大值
        }
        return R;
    }
};
```

Result

475. Heaters

Solved

Medium

Topics

Companies

Winter is coming! During the contest, your first job is to design a standard heater with a fixed warm radius to warm all the houses.
Every house can be warmed, as long as the house is within the heater's warm radius range.
Given the positions of `houses` and `heaters` on a horizontal line, return the *minimum radius standard of heaters* so that *those heaters could cover all houses*.
Notice that all the `heaters` follow your radius standard, and the warm radius will be the same.

Example 1:
Input: `houses = [1,2,3], heaters = [2]`
Output: `1`
Explanation: The only heater was placed in the position 2, and if we use the radius 1 standard, then all the houses can be warmed.

Example 2:
Input: `houses = [1,2,3,4], heaters = [1,4]`
Output: `1`
Explanation: The two heaters were placed at positions 1 and 4. We need to use a radius 1 standard, then all the houses can be warmed.

Example 3:
Input: `houses = [1,5], heaters = [2]`
Output: `3`

Constraints:

- $1 \leq \text{houses.length}, \text{heaters.length} \leq 3 \times 10^4$
- $1 \leq \text{houses}[i], \text{heaters}[i] \leq 10^5$

2.4K

89

13 Online

1

2

3

4

5

6

7

8

9

10

11

12

13

Accepted

Runtime: 0 ms

Case 1

Case 2

Case 3

Input

houses =
[1,5]

heaters =
[2]

Output

3

Expected

3

Contribute a testcase

475. Heaters

Solved

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Runtime: 0 ms

Case 1

Case 2

Case 3

Input

houses =
[1,2,3,4]

heaters =
[1,4]

Output

1

Expected

1

Contribute a testcase

The screenshot shows a web browser with the LeetCode problem '475. Heaters' open. The problem description is on the left, and the code and test results are on the right.

Problem Description:

475. Heaters

Medium Topics Companies

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Code:

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    for (int house : houses) {
        int minD = INT_MAX; // 將最小距離 minD 賦予 int 的最大值，方便找最小
        for (int heater : heaters) {
            minD = min(minD, abs(house - heater)); // 找最小距離並更新
        }
        R = max(R, minD); // 更新最大值
    }
    return R;
}
```

Testcase: Accepted Runtime: 0 ms

Case 1 Case 2 Case 3

Input

houses =
[1,2,3]

heaters =
[2]

Output

1

Expected

1

Discussion

我想到的做法是計算其中一個 house 與其最小距離的 heater，然後再從所有的 house 的最小距離中取得最大值，其餘的 houses 也會被覆蓋，這個最大值就是 R。

Lab01-Q#. (題目編號)

Briefly describe the problem. (題目簡述)

Code

Paste your code here. Please use [Pygments](#) to highlight your code.
(請使用 [Pygments](#) 轉換程式碼格式後再貼上)

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(請使用 [Pygments](#) 轉換程式碼格式後再貼上)

result

(a)

(b)

(c)

Show your result in tables or screenshots.
(請使用表格或是截圖呈現結果。)

Discussion

Discuss and conclude your results, or answer questions.
(實驗討論、結論或回答問題。)

[EXAMPLE] Q1. Factorial Function

The factorial function $n!$ has value 1 when $n \leq 1$ and value $n \cdot (n-1)!$ when $n > 1$.

Please write both a recursive and an iterative C function. Indicate the input, output, and give proper comments. Test your codes and log the results.

Code

<iterative version>

```
#include<stdio.h>
#include<stdlib.h>
int main()
{
    int n, answer;
    char i;
    printf("Factorial function n!, n=");
    scanf("%d", &n); // 輸入要查詢的 n
    if (n <= 1)
    {
        printf("Answer=1"); // 小於 1 時答案為 1
    }
    else // 大於 1 時
    {
        answer = n; // 計算 (n > 12 時超出 int 儲存上限)
        for (i = (n-1); i > 1; i--)
        {
            answer = answer * i;
        }
        printf("Answer=%d\n", answer);
    }
    system("pause");
    return 0;
}
```

<recursive version>

...

result

```
Factorial function n!, n=7
Answer=5040
```

Discussion

